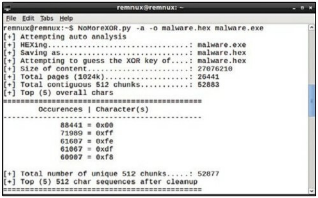

Figure 1: NoMoreXor

Figure 2: NoMoreXor for extracting possible XOR keys in the sample 'malware.exe'

Balbuzard

Balbuzard is a malware analysis tool that is used to extract patterns from malicious files and to crack obfuscated code using XOR. It is also used to extract strings/patterns and embedded files.

The syntax is:

```
Balbuzard.py  [option] [filename.extension]
```

Here's an example:

```
Balbuzard.py   --csv malware.csv malware.exe
```

Bbcrack

Bbcrack is a supplementary tool used to brute-force XOR keys based on patterns of interest.

The syntax is:

```
bbcrack.py [option] [filename.extension]
```

An example is:

```
bbcrack.py -l 2 malware.exe
```

Bbharvest

Bbharvest is also a brute force tool used to extract XOR keys that are specifically targeted against single

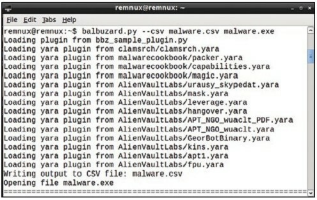

Figure 3: Balbuzard

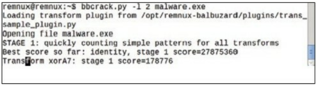

Figure 4: Bbcrack

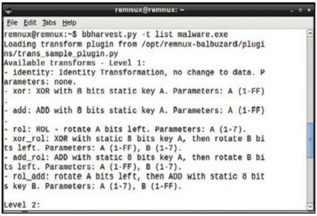

Figure 5: Bbharvest

obfuscated strings/patterns in malware.

The syntax is:

```
Bbharvest [option] [filename.extension]
```

Here's an example:

```
bbharvest.py -l 2 malware.exe
```

Bbtrans

Bbtrans is used to apply transforms over the malware samples.

The syntax is:

```
Bbtrans.py [option] [filename.extension]
```

And an example is:

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